MILITARY SPECIFICATION

TABLE, OPHTHALMIC LABORATORY, STEEL FRAME

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a work table for use in an ophthalmic laboratory for mounting lens surfacing equipment (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-E-527	Enamel, Alkyd, Lusterless
TT-P-636	Primer Coating, Alkyd, Wood and Ferrous Metal.
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Locked-Corner

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible

STANDARDS

Federal

FED-STD-66	Sheet, Chemical Composition and Hardenability
FED-STD-595	Colors

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Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications and standards, required by the suppliers in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

UNDERWRITERS' LABORATORIES, STANDARDS.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, L.I., New York 11749; 207 East Ohio Street, Chicago, Illinois 60611; or 1655 Scott Blvd., Santa Clara, Californis 95050.)

3. REQUIREMENTS

3.1 Fire and casualty hazards. The contractor shall submit to the contracting agency, prior to delivery, proof that the electrical components he will supply under this specification conforms to the Underwriters' Laboratories, Inc. The number of the U.L. Standard complied with shall be referenced in the document. The label or listing of the Underwriters' Laboratories, Inc., may be accepted as evidence that the equipment conforms to this requirement.

3.1.2 In lieu of the label or listing, the contractor may submit prior to delivery independent proof satisfactory to the contracting agency that his equipment conforms to the published standards, including methods of tests, of the Underwriters' Laboratories, Inc.

3.1.3 Compliance with Underwriters' Laboratories, Inc., requirements regarding fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his commodity.

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3.2 Materials.

3.2.1 Steel. Front and rear upright leg members, top and lower cross members, stringer, shelf, back and end stop shall be fabricated of hot-rolled carbon steel sheet. The receptacle strips shall be of suitable sheet steel, finished in gray paint. (see 3.3.3). All hardware used in assembling the table shall be of steel, suitable for the purpose intended.

3.2.2 Wood. The wood core for the table top shall be kiln-dried hardwood.

3.2.3 Laminated wood. Laminated wood for the table top shall be pressed wood.

3.3 Construction. The table shall consist of the top, two leg assemblies, a steel shelf and a stringer. The top shall have a back and end stop assembly, and be provided with an electrical strip containing five (5) evenly spaced receptacles. A six foot 3-conductor cable connected to the receptacle strip shall be supplied. The table shall be 60 inches long, 34 inches deep, and 34 inches high, when completely assembled. Table shall be supplied in a disassembled condition (see Fig. 1).

3.3.1 Top. Table top shall be 60 inches long, 34 inches deep, by 1-3/4 inches thick. The table top core shall be made of hardwood, laminated top and bottom with at least one layer of high density, tempered pressed wood, The table top shall be resistant to impact, moisture, and oils. The table top shall have pilot holes 1/4 inch in diameter by 1 inch deep for mounting the legs. Pilot holes shall also be supplied for attaching the back and end stop assemblies and the electrical outlet strip.

3.3.2 Back and end stop. The back and end stop assembly shall consist of a back board and two end stops bolted together. The back shall extend 3 inches above the table top. Each end stop shall taper from 3 inches at the back to 1-1/2 inches at the front, above the table top. End stops shall be not less than 28 inches. The assembly shall be of steel, 0.0747 inches thick (U.S.S. Gage No. 14).

3.3.3 Electrical outlet strip. An electrical outlet strip, for mounting under the front edge of the table top, shall be supplied. The strip shall be 48 inches long and have five grounded receptacles. The wiring for the strip shall be No. 12 gage. The strip shall be closed at one end with a blank end fitting, and the other end shall have an entrance end fitting. The electrical outlet strip shall be tested for continuity in accordance with 4.4.2.

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3.3.3.1 Interconnecting cable. A cable for connecting to the outlet strip shall be supplied. The cable shall be a 3-conductor, No. 12 copper stranded rubber or neoprene insulated cable. The cable shall be 6 feet in length, with a standard 3-prong male plug at one end. The other end shall be connected into the entrance end fitting of the junction box of the receptacle strip. The cable shall provide for electrical grounding.

3.3.4 Leg assemblies. Two leg assemblies, for mounting to the underside of the table, shall be supplied. Each leg assembly shall have two upright leg members, one front and one rear, a top cross member, and a lower cross member. Each leg assembly shall be completely fabricated by welding. The upright leg members, and cross members, shall each be made from one piece of heavy channel steel, double thickness "T" shape in cross section, accomplished by folding the channel through the center of its back. The completed leg assemblies shall be of such length that the top surface of the legs on which the table top is placed shall be 32-1/4 inches from the floor. The leg assemblies shall be 32-1/4 inches high by 34 inches deep.

3.3.4.1 Upright leg members. The upright members, after forming, shall be 2-13/16 inches wide across the arm, by 2 inches deep. The upright leg members shall flare out below the lower cross member, but shall not extend beyond the table top. The upright leg members shall have holes located as follows: Two inches up from the top surface of the lower cross member; then 4 inches up from the center of the first hole; then 6 inches up from the center of the second hole. Up from the center of the third hole, there shall be a series of four holes on 2 inch centers. The holes shall be in each flange of both upright leg members. The corresponding holes in the flanges of the leg members, shall be in the same horizontal line, and shall be 2 inches apart on centers. The metal of the uprights shall be 0.0747 inches thick (U.S.S Gage No. 14).

3.3.4.1.1 Base plates. Each upright leg member shall have a base plate welded to it. The plate shall be of such size as to provide 10-1/2 square inches of floor bearing surface per leg. The plate shall have two holes for attachment to the floor. The metal of the base plate shall be 0.0747 inches thick (U.S.S. Gage No. 14).

3.3.4.2 Lower cross member. The lower cross member shall be 2-13/16 inches wide by 1-1/2 inches deep. The cross member shall have holes on 3 inch centers in each flange. The corresponding holes in the flanges of the leg members shall be in the same horizontal line, and shall be 2 inches apart on centers. The cross member shall be welded to the front and rear vertical members. The top surface of the lower cross member, in the completed leg assembly shall be 9-1/2 inches from the floor. The cross member shall be of 0.1046 inches thick steel (U.S.S. Gage No. 12).

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3.3.4.3 Upper cross member. The upper cross members shall be 2-7/8 inches wide by 1-27/32 inches deep. The member shall have holes on 3 inch centers in each flange. The corresponding holes in the flanges of the leg member shall be in the same horizontal line, and shall be 2 inches apart on centers. The top member shall be of 0.1046 inch thick steel (U.S.S Gage No. 12). The cross member shall be welded on top of the vertical members. The top surface of the upper cross member, in the completed leg assembly, shall be 32-1/4 inches above the floor.

3.3.5 Stringer. A stringer, connecting the rear upright legs, shall be provided. The stringer shall be channel type 59-7/8 inches long, 5 inches wide, and have at least four holes on each end, to permit attachment of the stringer to the uprights, by means of nuts and bolts and four matching holes in the flanges of each leg. Steel shall be 0.0478 inches thick (U.S.S Gage No. 18).

3.3.6 Shelf. A shelf 60 inches long by 13-1/4 inches wide shall be supplied. The shelf shall be flanged on all sides. The rear flange shall be 1 inch wide, shall be vertical upward, and shall be provided with holes for attachment to the rear upright members of the two legs. The two side flanges shall be vertical downward. The front flange shall be a reverse flange with holes for attachment to the lower cross member. The shelf shall be 0.0598 inches (U.S.S Gage No. 16) in thickness.

3.3.7 Hardware. Suitable steel nuts, bolts, washers, and lug screws shall be supplied for assembling the table in the field.

3.4 Finish. All accessible metal parts, except the electrical parts, shall be thoroughly cleaned and chemically degreased. After degreasing, a primer coat shall be applied using a primer conforming to TT-P-636 and two coats of lustreless olive drab enamel conforming to TT-E-527. The color shall conform to color 34087 per FED-STD-595 (see 6.4). The enamel shall be sprayed on and baked in accordance with the paint manufacturers recommendations. The final coat of enamel shall level out to produce a smooth, hard surface, resistant to abrasion, chipping, and cracking.

3.5 Identification marking. Each table shall be permanently and legibly marked in black on one of the front upright leg members with the name or registered trademark of the manufacturer and the contract number. As an alternate, the name or registered trademark of the manufacturer may be embossed in the steel.

3.6 Instruction for assembly. The table shall be supplied in a knockdown condition. Complete instructions shall be provided for assembly of each unit in the field.

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3.7 Workmanship. Workmanship shall be first class throughout. The table shall be free from defects which detracts from its appearance or may impair its serviceability. Table shall be free from burs and sharp edges. It shall be entirely suitable for heavy-duty service in ophthalmic laboratories.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I.

Table I. Sampling for examination

Examination	Inspection level	AQL (percent defective)
Major	II	1.0
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105.

4.3 Examination.

4.3.1 Table. The table components shall be examined for defects including, but not limited to, those listed in Table II. Minor B defects shall be performed on an assembled table.

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Table II. Classification of defects

Major	Defect
101	Any component missing or not as specified.
102	Materials nonconforming to specification.
103	Table not free of burrs, sharp edges or splinters on exterior edges.
Minor A	
201	Welds not ground smooth.
202	Welds exhibit cracks or porosity.
203	Identification marking missing or incorrect.
Minor B	
251	Assembly hardware missing or of wrong size.
252	Holes for assembly missing or improperly located.
253	Overall dimensions out of tolerance.

4.3.2 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table III. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

Table III. Examination for preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

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4.4 Tests. Tests of the materials used in the fabrication of the tables shall be conducted as specified herein. Other tests may be made, as necessary, to determine that the finished tables comply with this specification, and that the materials comply with the applicable portions of the specifications and standards listed in section 2.

4.4.1 Continuity test. The unit shall be plugged into a nominal 115 volt, AC power source. Each outlet shall be tested by means of a test lamp fitted with a standard male plug. Upon insertion of the test lamp in each outlet, the lamp shall remain lighted without failure. Failure of the lamp to illuminate in any outlet shall be cause for rejection. Continuity shall be tested from the ground prong of the male plug to the ground contact of each receptacle.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each table shall be disassembled and all components wrapped in kraft paper and cushioned with corrugated fiberboard or other suitable material where required. Wrapping and cushioning shall be secured with tape. As an alternate, components may be packaged in fiberboard boxes.

5.1.2 Level C. Each table shall be packaged in accordance with standard commercial practice which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Each table, packaged as specified in 5.1, shall be packed in an exterior container designed for a type 3 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type or PPP-B-621, class 2. The unit shall be adequately secured within the exterior container by means cushioning, blocking and bracing to insure against shifting and damage. Closure and strapping of the container shall be as specified in the appendix of the applicable box specification. In addition, the exterior container shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing of the liner shall be as specified in the liner specification.

5.2.2 Level B. Each table, packaged as specified in 5.1, shall be packed in an exterior container designed for a type 3 load and contructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type or PPP-B-621, class 2. The table shall be secured within the exterior container as specified in 5.2.1. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. Each table shall be packed in a substantial commercial container of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking. Exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The table covered by this specification is intended for use in an ophthalmic laboratory to accommodate NSN 6540-00-720-9683, Ophthalmic Lens Surfacer.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(c) The National Stock Number (NSN).

6.3 This specification is intended to cover the following item listed in the Federal Supply Catalog:

NSN	Item Identification
6540-00-721-9637	Table, Ophthalmic Laboratory, Steel Frame

6.4 Availability of color chips. The standard color chip of the color stated in 3.4 may be obtained from:

Commissioner
Federal Supply Service
General Service Administration
Standardization Division
Paints Branch
Washington, DC 20406

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6540-1039
cs

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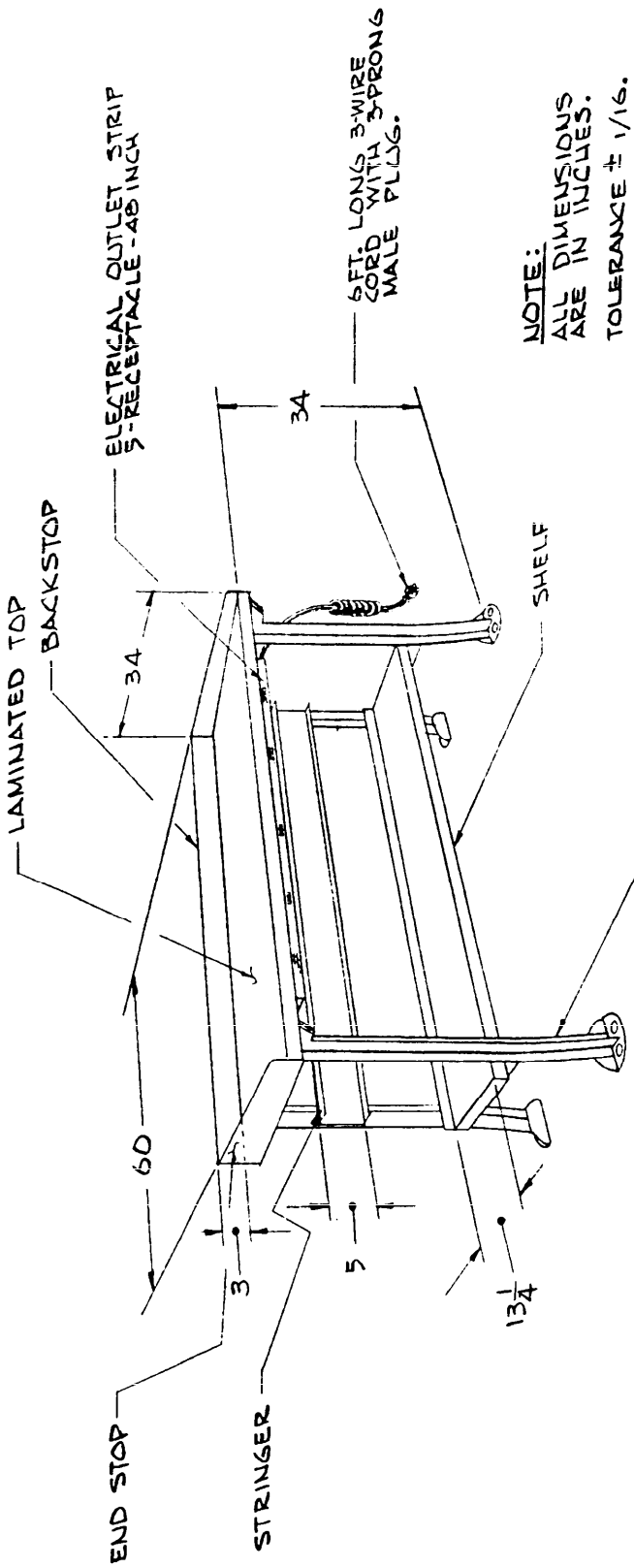


FIGURE 1 - TABLE, OPHTHALMIC

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DOCUMENT IDENTIFIER AND TITLE

MIL-T-37188 TABLE, OPHTHALMIC LABORATORY, STEEL FRAME

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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